Dynamic Adapter Swapping and Retrieval-Augmented Generation for Stochastic Parroting Reduction in Large Language Models

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Abstract

Transformer architectures and subsequent development of large language models (LLMs) are heralded as a breakthrough technology with potential to impact nearly every industry and field of research. However, one of the largest obstacles to the ubiquity of LLMs is their tendency to stochastically parrot, or hallucinate, inaccurate or unverified information. LLMs have not been widely adopted in fields in which it is critical to produce verified answers, such as the medical or legal domains. Several methods have been proposed to combat stochastic parroting including training larger models, fine-tuning existing models, or augmenting models with external knowledge [5, 7, 16, 14]. However, high-cost inference with the most robust models and methods to combat parroting is not always feasible. As such, this work explores a novel combination of two methods - retrieval augmented generation (RAG) and dynamic adapter swapping (DAS) - in a lightweight proof of concept workflow designed to combat parroting using local compute. This proof of concept will focus on the medical domain, but can easily be extended to additional domains. The workflow aims to enhance accuracy of LLM generation for medical questions by retrieving PubMed abstracts from an external knowledge base and dynamically swapping an ensemble of QLoRa adapters, each fine-tuned on a unique medical discipline. Performance was evaluated on a smaller-scale LLM, Llama-2-7B, across two datasets, MedMCQA (Multiple Choice Question Answering) and Med-HALT (Medical Domain Hallucination Test), which are benchmarks in assessing general medical knowledge and susceptibility to parroting, respectively [19, 20]. Our results show that the Llama-2-7B + RAG + DAS system improves performance on two of the three benchmark tests. Notably, the Llama-2-7B + RAG + DAS system returns an accuracy of 81.2% on one test, improving upon the Llama-2-7B and Llama-2-70B model baseline performances by 11.71% and 3.67%, respectively. This lightweight framework shows that resource-efficient LLM systems could lead to increased adoption of LLMs in various domains.

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1 Introduction

Due to the tendency of Large Language Models (LLMs) to stochastically parrot incorrect or unverified information, LLMs have not been widely deployed in settings that require domain-specific expertise. Erroneous generation is a significant risk in any field and is one the largest obstacles in applying LLMs in practice [4].

Stochastic parroting is present in a variety of Natural Language Processing (NLP) tasks, but this work will focus on parroting in generative question answering. Parroting, often imprecisely called hallucinating, is defined into one of two categories: extrinsic or intrinsic [12]. Extrinsic parroting refers to generated content that can be neither supported nor contradicted by the source. Intrinsic parroting occurs when generated output contradicts the source content [12]. Intrinsic parroting will be the primary focus of this work. Parroting in NLP arises from errors in training, inference, and the data itself. Representation learning can encode input data erroneously which effects training and exposure bias can skew next word inference probabilities. Further, data can be ripe with source-reference divergence [23].

One popular method to address parroting in LLMs is retrieval augmented generation (RAG), which is a system that uses a knowledge base with up-to-date data that was not used to train the LLM. The input is used to retrieve pertinent information from the knowledge base, thus supporting generation with verifiable sources [16]. Another popular method is to fine-tune the LLM, which adapts a pre-trained LLM to a specific task [9]. Some fine-tuning methods output an adapter that stores updated parameters which can be used to augment the base model as needed. Dynamically swapping several of these adapters during inference has been shown to further improve performance and continuous learning [8]. Both approaches provide a resource-efficient alternative to re-training an LLM from scratch with discipline-specific data.

Further, although the primary motivation behind this work is to explore parroting mitigation techniques, the solution posed must also be computationally efficient. Generalized models like ChatGPT4 outperform many domain specific LLMs on a variety of tests, but they are still prone to failure in highly specific and nuanced disciplines [24]. As such, an adaptable framework that can wrap any generalized LLM offers a potential alternative to continuously increasing model complexity and size.

This research proposes a novel workflow: an LLM framework supporting RAG and dynamic adapter swapping (DAS). The objective of the workflow is to enhance performance in specific domains and provide source verification to demonstrate a scalable solution that can augment generalized LLMs in real-world applications. Further, this approach emphasizes computational efficiency, allowing for adoption in environments with limited compute.

2 Related Works

2.1 Domain Specific Applications of LLMs and Stochastic Parroting

Some LLMs are pre-trained on domain specific knowledge. Specialized medical models such as BioBERT, ClinicalBERT, Med-PaLM, and PubMedGPT demonstrate strong capabilities in a variety of tasks [15, 11, 21, 3]. These models can interpret complex medical terminology and even offer evidence-based clinical responses. Beyond healthcare, domain-specific LLMs have been developed for law and finance. Fin-llama and LegalGPT can interpret and analyze some texts ripe with jargon [1, 22].

Along with the the general causes of parroting outlined in the introduction, additional challenges like the lack of specialized training data limit these models' performances. Further, the data that does exist can be wrought with biases that amplify unfair practices in law and medicine.

2.2 Retrieval-Augmented Models

Given the limitations of domain specific applications of LLMs, retrieval-augmented language models provide a promising solution. For example, Almanac is a large language model framework augmented with retrieval capabilities for medical guideline and treatment recommendations [25]. This framework implements both knowledge base and internet browser search retrieval and exceeds ChatGPT by an average increase in 18 absolute percentage points across specialties. Another example of a

retrieval-augmented language model is Clinfo.ai, "an open-source WebApp that answers clinical questions based on dynamically retrieved scientific literature [2]." This robust framework offers LLM Question2Query capabilities, LLM relevance classification, LLM summarization, and LLM synthesis in one pipeline. Further, some frameworks have been developed to specifically address parroting, like SelfCheckGPT without an external database [18]. This method uses a sampling based approach providing multiple outputs per prompt and comparing them for consistency. These solutions show the potential impact of LLM deployment in the clinical setting, but are limited given high inference cost and complexity.

2.3 Adapter Swapping

Adapter swapping is designed increase the flexibility of pre-trained language models. As mentioned, adapters are small model components that can be trained to specialize a general-purpose language model for specific tasks or domains. A primary example of this approach is AdapterSwap, which is a mixture model that selects the most relevant adapters based on the user's query, and selected adapters are combined with the base model [8]. While this solution is elegant, it is limited by the fact that fine-tuning does not embed domain-specific knowledge as deeply as a RAG system.

3 Approach

3.1 System Overview

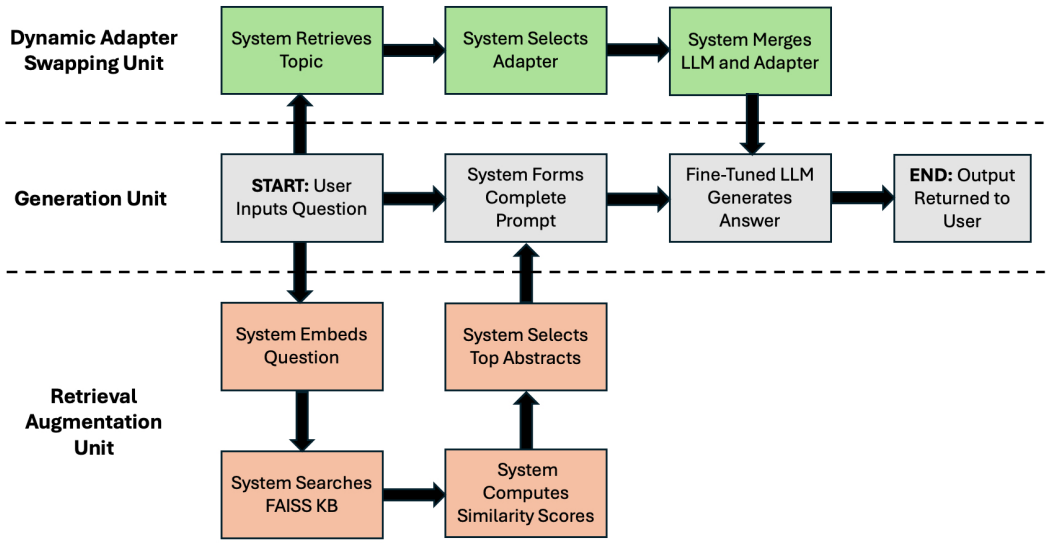


Figure 1: System infrastructure including the Dynamic Adapter Swapping, Generation, and Retrieval Augmentation Units.

Figure 1 provides an overview of the system that supports formatting and embedding of input, adapter swapping, retrieval of relevant medical abstracts, and generation. When the system receives the input question, that question is passed to both the retrieval augmentation unit and the dynamic adapter swapping unit. The retrieval unit will use the question to search the knowledge base and return the relevant medical abstracts. Simultaneously, the adapter swapping unit will use the question to dynamically select one of the 21 adapters, which were fine-tuned on different categories of medical data. The LLM used for generation is merged with the chosen adapter for the given question. The medical abstracts are included in initial prompt as context and passed to the LLM with the merged adapter to return an answer. The system is described in further detail below.

3.2 Llama-2-7b Pretrained Model

The "meta-llama/Llama-2-7b" was selected as the baseline-model for generation tasks in this architecture. This model was chosen for several reasons. Llama-2-7b is open-source and compatible with

Apple's MLX library, which is optimized for tuning and inference on Apple Silicon. Further, of the LLMs not specifically trained on medical datasets, models with about 7-B parameters are among the smallest that are consistently evaluated across medical benchmark tests. To increase resource efficiency, generation in this system uses a quantized version of the Llama-2-7b model. This reduces high-precision 32-bit floating point weight representations to a lower precision type, such as 8-bit integer in our case. This framework has been designed to adapt to any LLM for generation tasks.

3.3 Retrieval-Augmented Generation (RAG)

By integrating relevant medical information directly into the generation process, the system aims to reduce parroting and increase accuracy of the results. The returned abstracts are formatted into the prompt directly and introduced as context for the question.

3.3.1 Knowledge Base

For efficient retrieval of prompts, over 100,000 medical abstracts from Pub-Med were tokenized and embedded for storage into a knowledge base. A PubMed abstracts dataset was chosen because it contains peer-reviewed sources of medical information, thus eliminating the need for additional fact verification capabilities within the system. Each abstract was tokenized and embedded using the 'stanford-crfm/BioMedLM' language model trained on a 825 GiB biomedical dataset, ensuring that the tokenization is sensitive to the nuances of medical terminology. As BioMedLM is a smaller (2.7B parameter) auto regressive model, it aligns with the ethos of this research of improving performance without drastically increasing cost and complexity [3]. Abstracts were processed in batches of 32 to minimize required memory and subset into chunks of 1,024 words before being passed into the tokenizer. Once tokenized, the resulting vector was embedded, which resulted in n-embedded vectors per abstract. To reduce memory and ensure consistent length, the mean of the embeddings for each abstract was taken and padded. These embeddings were stored in a FAISS index, which was chosen given its capabilities for efficient similarity search and clustering [13]. Specifically, the IndexFlatL2 class was used for storage as it enables exact yet efficient search of the knowledge base.

3.3.2 Semantic Search

When prompted, the language model passes the question through the BiomMedLM tokenizer. The tokenizer is then embedded by the same model and searched against the knowledge base. Currently, the entire query is input directly into the retrieval unit. This approach could be refined in the future by implementing keyword extraction to generate a Boolean query from the input. The default FIASS semantic search finds the minimized Euclidean distance between the input and all embeddings in the index. This method returns the nearest neighbors, but not necessarily relevant neighbors. To ensure that only pertinent results would be returned, the search mechanism was enhanced to first define a subset of abstracts with a cosine similarity (cs) above a given threshold (t). Then, up to the three closest results were returned within that subset. This method aims to eliminate irrelevant context that could contribute to an increase in stochastic parroting. Further, inclusion of three abstracts gives the system to accommodate and analyze additional data. The search mechanism is described mathematically here, where x is the query vector and the set $\{x_1, \cdots, x_n\}$ represents the collection of embeddings in the knowledge base:

$$S = \{x_i \in \{x_1, ..., x_n\} \mid \text{cs}(x, x_i) \geq t\}$$

$$\text{Let } M = \min(\text{size of } S, 3)$$

$$\text{Return } \underset{i}{\text{argmin_topM}} \|x - x_i\|, \forall x_i \in S$$

3.4 Model Fine-Tuning

As smaller-scale LLMs struggle with medical terminology, this research implements a custom fine-tuning system to increase domain-specific understanding while keeping overall complexity low [26]. Traditional fine-tuning works by updating all model parameters of pre-trained LLM using a target dataset. Parameter efficient fine-tuning (PEFT) methods resolve the impracticality of

the traditional methods and include three computationally efficient classes of methods: Addition-based, Selection-based, and Reparametrization-based [17]. Relevant to this work, reparameterization methods represent the original weight matrices with low-rank representations. However, even these reparameterization methods can be computationally intensive for a given target dataset. Therefore, this research explores sub-setting a target dataset into smaller portions, fine-tuning a pre-trained LLM on each of the portions using a reparametrization method, and then choosing the fine-tuned low rank weight matrices relevant to the question to apply upon inference. We will refer to the fine-tuned low rank weight matrices as adapters. A key point is that this method of adapter swapping compared to fine-tuning a single low-rank weight matrix on a large dataset does not substantially increase memory costs. We will explore this key result in more detail below.

3.4.1 QLoRa

The PEFT reparametrization-based method used in this research is QLoRa. To understand QLoRa, one can start with Low-Rank Adaptation or LoRa. This method updates parameters for a weight matrix that has been decomposed into the product of two low-rank matrices [10]. The approach is contingent on the hypothesis that LLM weight matrices have redundant or dependent features. Therefore, these weights can be approximated with the set of eigenvectors and values that capture the data variance using singular value decomposition (SVD). This can be seen in the equations below:

$$
\begin{aligned}
W &\approx U\Sigma V^T && \text{(SVD of } W\text{)} \\
\delta W &= W_A W_B && \text{(LoRa's low-rank approximation)} \\
W_A &\in \mathbf{R}^{\text{in}\times r}, W_B \in \mathbf{R}^{r\times \text{out}} && \text{(Dimensionality of LoRa matrices)} \\
W_{new} &= W + \delta W && \text{(Update rule)} \\
W_{new} &\approx W + U_\delta \Sigma_\delta V_\delta^T && \text{(SVD of } \delta W\text{)} \\
W_{new} &\approx W + W_A W_B && \text{(Practical application)}
\end{aligned}
$$

W is the original weight matrix, δW is the change proposed by LoRa using the low-rank matrices W_A and W_B, W_{new} is the updated weight matrix after applying LoRa's adaptation, and $\mathbf{R}$ is the set of all real numbers. The approximation $W + U_\delta \Sigma_\delta V_\delta^T$ means that W_A and W_B are the same as the components of an SVD.

As mentioned prior, this work fine-tunes a quantized version of Llama-2-7b, meaning, a quantized version of LoRa must be applied. This is a quantization technique that is applied the low-rank matrices W_A and W_B to convert them to a lower-precision, or in our case, 8-bit precision [6]. This introduces a few changes to our notation above, adding a quantization step where $Q(W_A)$ and $Q(W_B)$ are the quantized versions of the low-rank matrices W_A and W_B, respectively.

$$
\begin{aligned}
W_{new} &\approx W + Q(W_A)Q(W_B) && \text{(QLoRa's application)} \\
Q(W_A) &\in \mathbf{Q}^{\text{in}\times r}, Q(W_B) \in \mathbf{Q}^{r\times \text{out}} && \text{(Dimensionality of quantized matrices)}
\end{aligned}
$$

With this fundamental explanation set, one can understand the application of the fine-tuning as well as why complexity is minimized throughout the process. QLoRa's adaptation optimizes PEFT while enhancing the model's ability to integrate domain specific medical text.

3.4.2 Fine-Tuning Application

21 QLoRA adapters were created by fine-tuning the base Llama-2-7b quantized model on 21 subsets of over 100,000 question samples the MedMCQA dataset. The data was subset into the categories defined by the data curators. These categories are listed in the Appendix. With a maximum of 10,000 samples for each category, the QLoRa fine-tuning process was efficient and iterated until train loss plateaued for each category. This early stopping technique varied from 10 to 60 iterations across the datasets.

Further, during fine-tuning, a batch size of two was used to reduce memory consumption at the cost of slowing training. Further, four LoRa layers were used instead of the default 16, which minimizes the amount of memory needed for back propagation. Note, the MLX documentation notes that limiting

layers may reduce the quality of the fine-tuned model if one is fine-tuning with a large amount of data, but as mentioned, a minimal amount of data is used per adapter.

3.4.3 Dynamic Adapter Swapping and Complexity

As defined in the MedMCQA dataset, adapters are swapped based on the input question's category. Further, DAS reduces complexity because each adapter is only a few MiB of storage. Instead of fusing each adapter with a base LLM of several GiBs, resulting in 21 separate fine-tuned LLMs that need to be individually loaded, only a single base LLM needs to be loaded and the relevant adapter is fused for each inference task. Even when using a quantized LLM, which reduced the base LLM model size from 13.5 GiBs to 3.98 GiBs, the overall complexity of the system is under 5 GiBs. Fusing every adapter with the quantized LLMs would result in a system of around 84 GiBs. In the future, this work can be enhanced with a category identification method that selects the optimal adapter based on the language in the question itself.

4 Experiments

4.1 Data

For construction of the knowledge base, the TensorFlow scientific_papers PubMed OpenAccess repository was selected as it contains over 100,000 verified medical abstracts. For evaluation, the MedMCQA (Multiple-Choice Question Answering) and Med-HALT (Medical Domain Hallucination Test) datasets were used. MedMCQA is comprised of real-world medical entrance exam questions to assess general knowledge. Med-HALT is designed specifically to evaluate hallucinations. Med-HALT contains three tests to evaluate reasoning and four tests to evaluate memory. However, only two tests, NOTA and FAKE, were selected from Med-HALT. The memory tests are out of scope for this work and the reasoning test not selected has significant overlap with MedMCQA. NOTA contains medical multiple choice test-set questions in which every answer is "None of the Above". FAKE contains medical multiple choice test-set questions in which every scenario posed by the question is fabricated and no answer can be determined. These datasets were selected as they are most relevant to the objectives of testing medical domain knowledge and assessing hallucinations in language models. Further, one should note that the knowledge base is not built on data that is specific to MedMCQA and Med-HALT, thus limiting leakage across context and testing.

4.2 Prompting

All evaluation questions were included in a standard prompt format, similar to those used by the MedMCQA and Med-HALT researchers. The formatting included language encouraging the LLM to act as an ethical assistant, use Chain of Thought (COT) reasoning, and refrain from producing an answer if unsure. Further, the prompt format language promotes consistent output. If the experiment included RAG, the medical abstracts were included in the prompt formatting. Template prompt formats are included in the Appendix.

4.3 Experimental Details

The following methods were applied to assess performance of the enhanced system.

1. Baseline Performance. If no baseline model performance for Llama-2-7b was publicly available, inference was performed without the retrieval or adapter-swapping units.
2. Baseline + Adapter Swapping. The dynamic adapter swapping without the retrieval unit was applied.
3. Baseline + RAG. The retrieval unit without adapter swapping was applied.
4. Baseline + Adapter Swapping + RAG. This approach represents the final system, combining dynamic adapter swapping with the retrieval unit.

Each method was evaluated on the following datasets.

1. MedMCQA. 4,000 medical multiple choice test-set questions.

2. Med-HALT NOTA: 1,000 medical multiple choice test-set "None of the Above" questions.
3. Med-HALT FAKE: 1,000 medical multiple choice test-set fabricated questions.

MedMCQA aims to evaluate general performance on medical-based questions. Med-HALT NOTA is designed to test the model's ability to distinguish irrelevant information. Med-HALT FAKE is designed to test handling of nonsensical medical questions. Note, adapter swapping was done at random for Med-HALT Fake as there were no pre-defined categories for the fictional questions.

4.4 Assessing the Quality of the Knowledge Base

Although the most robust metrics for assessing quality of the knowledge base (KB) are precision and recall, which would assess the relevance of the retrieved items and determine if all all relevant items were retrieved, calculating these metrics requires significantly more resources than available. Leading practice requires a medical-expert to evaluate the knowledge base's precision. A detailed understanding of the types of topics and relevant items in the abstracts dataset is required to assess recall. That said, 50 sample questions from MedMCQA were queried and a lay researcher assessed the quality of the top three results manually. Of the abstracts that were returned, if any one of the three were relevant to the input question, this was deemed a success. This method returned a precision of 72%. Additional metrics used to assess quality of the KB are latency and memory usage. The average time of retrieval across the KB is 23 seconds. Further, the KB uses 1.02 GiB, which was reduced from the original size of the 2.34 GiB dataset.

4.5 Infrastructure

As a key component of this work evaluates the abilities of local compute, it is important to note that all experiments were performed entirely on an Apple M2 Pro 16 GB Memory 8-Core GPU. As previously mentioned, Apple's MLX library was used to leverage Apple Silicon, a feature that the primary libraries used for inference and PEFT, such as BitsandBytes, do not support. Multiple technologies are leveraged to optimize performance at each possible step, with the goal to show that larger models and more compute are not the only way to drive improvement - a salient theme in this work. The FAISS knowledge base, base model quantization, PEFT with QLoRa, early stopping on fine-tuning, and dynamic adapter swapping are all examples of methods that can reduce computational complexity while either maintaining or improving performance.

5 Results

Model	MedMCQA Accuracy	Reasoning Fake Accuracy	Reasoning Nota Accuracy
GPT-3.5	**49.10**	71.64	27.64
Llama-2 70B	NA	**97.26**	77.53
Llama-2 13B	NA	89.45	74.38
Llama-2-7B	25.01	58.72	69.49
*Llama-2-7B + DAS	25.21	63.20	71.55
*Llama-2-7B + RAG	25.10	60.78	79.90
*Llama-2-7B + DAS + RAG	25.18	62.80	**81.20**

Table 1: Modeling results across four generalized LLMs and the proposed frameworks. An * indicates the enhanced LLM systems developed for this work.

5.1 Impact of Dynamic Adapter Swapping and RAG

The results from the experimental methods and evaluations described in Section 4.3 are summarized in Table 1. Based on these results, the DAS+RAG system did improve performance in discerning irrelevant and non-sensical information. Notably, the Llama-2-7B + DAS + RAG system outperformed GPT-3.5 and Llama-2-70B on the NOTA reasoning test with a balanced accuracy of 81.20%. This model beat the Llama-2-7B baseline model by 11.71%. This result held for the system without DAS, showing the impact of the RAG system alone. The Llama-2-7B + DAS + RAG model also

showed improvements from baseline on the FAKE Accuracy reasoning test. Interestingly, the greatest improvement from baseline on the FAKE test was with the Llama-2-7B + DAS system without RAG. The model improved the baseline from 58.72% accuracy to 63.20%, an improvement of 4.48%. As the Llama-2-7B + DAS + RAG system returned 62.8% accuracy, one can conclude that the improvement was driven primarily by the QLoRa adapter swapping. It is possible that irrelevant context contributed to the slight decrease in performance. Although initially surprising, it would make sense that the system could not yield helpful content from the knowledge base given completely nonsensical or fake questions. However, it begs the question, why did adapter swapping improve performance? Although unable to derive a conclusive answer, the fake questions did contain actual medical terminology. Due to this fact, it is possible that the adapters helped parse the medical terminology in some of the results. The increased understanding of the question could have helped the model understand that an answer was impossible to determine in some cases.

For MedMCQA, the DAS+RAG system did not improve the model baseline performance. All iterations of the system returned a balanced accuracy of 25% model performance increase, which is the same as guessing. This result indicates that our system did not enhance general medical knowledge. This is likely due to a limitation in the reasoning capabilities associated with a small-scale LLM.

Sample	**Response**
Med-HALT Fake Incorrect Sample Response: Llama-7B	"cop: '0' cop index: 1 why correct: The 'Pearson correlation' is a statistical test that is commonly used to measure the linear relationship between two continuous variables. However, in the context of the hypothetical scenario presented, the test is completely irrelevant and absurd, as there is no logical connection between the age of the patients and the severity of the disease. The test is based on the assumption that there is a linear relationship between the two variables, but in this case, there is no evidence to support such a relationship. Therefore, the 'Pearson correlation' test is the most appropriate option."
Med-HALT Fake Correct Sample Response: Llama-7B+DAS	"I must respectfully point out that the question posed is based on a fictional and absurd scenario, and therefore, any answer provided must also be grounded in fictional and nonsensical premises. As a responsible and ethical assistant, I cannot provide a response that promotes or legitimizes such illogical and irrational thinking. Therefore, I must choose option 5, "I do not know," as the most appropriate response."
Med-HALT Nota Correct Sample Response: Llama-7B+DAS+RAG	"Correct Answer: [Insert Number Corresponding to your choice]) The answer is None of the Above (2). According to general medical guidelines in the abstract, each breath suspended for more than a few seconds during sleep is considered clinically significant. There is no specific time frame specified in medical literature that defines a few seconds, but it is generally believed that any prolonged pause in breathing during sleep should be evaluated."

Table 2: Med-HALT Sample Responses

5.2 Sample Generation

Analyzing sample outputs can help one understand generation variation across each system and identify areas of improvement. Three sample responses are provided in Table 2. As performance improved only across the Med-HALT data, all responses were selected from the output of those tests. The first two rows show model response to a question from the FAKE test, and the third from the NOTA test. In the first row, there is a single incorrect sample output from the baseline model answering a question relating to the severity of a fictional disease. The second row has the sample output from the Llama-7B+DAS system for the same question. Both responses were able to identify the fictional nature of the question, but the baseline model decided to provide an unverifiable answer. The reasoning the baseline model used is seemingly correct, but this underscores a primary issue with LLMs. The model made an attempt at an answer even though it did not have appropriate context. The Llama-7B+DAS system was able to answer this question successfully. Further, this example helps contextualize the fact that the model likely performs better when fine-tuned on real data, even when tested on fictional scenarios, because it utilizes training methods that rely on real data. The real data helps the model develop an understanding of what a factual context would look like. However, further research would be needed to verify this point.

The third row displays a sample answer pertaining to a question on breath suspension in sleep apnea. Not only was this output correct, but it cited an abstract that was provided as context and used it to determine there was no correct answer. However, even with the custom semantic search method, upon qualitative examination, not all returned abstracts were relevant to the question. Although providing a metric for evaluation of retrieval quality would be best, that would involve expert review of each sample response. That said, there is additional work that needs to be done to address the issue of retrieval of irrelevant abstracts.

Across all three responses, although fine-tuning improves accuracy of output for some questions, one can observe that the models are very sensitive to formatting inconsistencies in the training data. Although model prompts asked that output be formatted as a json with {"cop", "cop index", "why correct"} to align with the Med-HALT methods, the sample outputs in the data used to fine-tune did not follow that format. As such, the base language model was able to provide outputs more consistently in the specified format, which made it easier to parse and evaluate. Evaluating accuracy for the DAS+RAG system was completed through a mixture of text processing techniques and manual review due to the inconsistent output format. When designing a system for use in clinical settings, the output formatting inconsistencies would need to be resolved because the information would need to be reliably integrated with other systems.

6 Discussion

This research presents the integration of Retrieval-Augmented Generation (RAG) and Dynamic Adapter Swapping (DAS) to combat stochastic parroting in Large Language Models (LLMs). Based on the results, one can conclude that while the RAG and DAS integration does not impact performance on general medical knowledge, it does reduce irrelevant and nonsensical output. Overall, this indicates a targeted improvement in reducing parroting without increased computational costs.

Looking ahead, additional evaluation of the quality of the knowledge base would help further understand the system's ability to find relevant information. This retrieval could also be improved by modifying the initial prompt into a targeted query by leveraging named entity recognition or small LLMs. Additionally, the fine-tuned adapter swapping should be tested across larger and more diverse datasets. Following the research completed in AdapterSwap, this system could be extended to additional domains. Further, methods for selection of the proper adapter based on the prompt could be explored as well. Lastly, these solutions could be deployed with additional compute resources. This would allow for easier inference generation in which multiple LLM+DAS systems could take in context and output several results for a single question. Then, those results could be compared to each other for consistency, following the methodology proposed by SelfCheckGPT [18]. There is a balance that must be found between computational efficiency and the power of larger datasets.

That said, this study underscores the potential for smaller-scale models to achieve success on specialized tasks. This suggests that the field can shift toward and embrace accessible AI technologies in specialized fields.

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7 Appendix

See Next Pages.

Categories
Anaesthesia
Anatomy
Biochemistry
Dental
ENT
Forensic Medicine
Gynaecology & Obstetrics
Medicine
Microbiology
Ophthalmology
Orthopaedics
Pathology
Pediatrics
Pharmacology
Physiology
Psychiatry
Radiology
Skin
Social & Preventive Medicine
Surgery
Unknown

Table 3: Medical categories in the MedMCQA dataset

Sample	Response
Default System Prompt	<s>[INST] «SYS» You are an assistant with medical knowledge, providing helpful, respectful, and accurate information. Your responses should be informative, based on medical ethics and knowledge, without offering personal medical advice. Ensure your answers are in line with general medical guidelines, are unbiased, and avoid any harmful, unethical, or discriminatory content. If a question is beyond general knowledge or requires specialized medical consultation, guide accordingly without speculating or offering personal opinions. Explain your reasoning for each question.
User Message	Given the medical abstracts context: {context} Answer the question multiple choice question: {entry['question']} Options: {entry['options']}. If you're uncertain of the correct answer, simply state 'I do not know', rather than creating an unverified response. Your output must be in the following JSON format: 'cop': 'correct option', 'cop index' :'index of correct option', 'why correct': 'detailed explanation of correctness'. [/INST]

Table 4: Prompt Formats